

**Analyzing the Effect on Life Expectancy Using the Key
Economic Indicators - Final Report**

CS 17600: Data Engineering in Python

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Group Members

Niharika Rai – rai55@purdue.edu

Rachel Carlson – carls171@purdue.edu

Nyree Ruiz – ruiz165@purdue.edu

Elia Lies – elies@purdue.edu

Introduction

Our main aim is to predict the effect on life expectancy in males and females using key economic indicators in any given country such as healthcare expenditure, GDP per capita, and the Gini Coefficient. In our datasets, healthcare expenditure showed the amount of spending done by a country on healthcare facilities. GDP per capita showed the amount of income generated per person in each country. This helped us see the standards of living of an individual living in that country. And lastly, the Gini coefficient showed the level of inequality in life span between men and women in each country. Through our data-wrangling processes, we were able to achieve our aim and demonstrate how each of these key economic indicators independently affected the world life expectancy and the male and female life expectancies over a span of 20 years in the 21st century.

Data Wrangling Process

Cleaning

We cleaned the data by getting rid of the rows of data that had missing values, because without these values we would not be able to filter or graph as efficiently as we would have. However, we could have also filled in the missing values using a variety of ways, but we decided that the number of missing values that were dropped would not have impacted on our findings or results. As previously mentioned, we decided that it would be best to remove the rows that had missing row values from the dataset because they would inhibit us from being able to complete the visualization that would be the most useful. To do this we decided to start off with using a for loop which goes through each of the files and replaces the empty row values with an NA to make it easier to spot where the missing or empty data values are in the data files. That code is attached below.

```
[20]: #Replace null or blank values with NA
      for df in [spending_life_expectancy, life_exp_inequality]:
          df.replace(['', ' ', 'NA', 'null', '?'], pd.NA, inplace=True)
```

Following that step we used a series of `dropna()` to remove the rows that had missing values. This process allowed us to ensure that we have identified all the missing data by using the "for loop" to identify whether it is a question mark or a blank space. In addition, using the `dropna()` was the most efficient and effective way of removing the rows with the missing data. That code is attached below.

```
[26]: #drop rows with null values
      spending_life_expectancy.dropna(inplace=True)
      life_exp_inequality.dropna(inplace=True)
```

Filtering

We ran into some problems with the filtering section. Originally, we were going to filter the data so that we only had datapoints in the range from 2014-2024. Then we were going to keep the most recent year for each entity and drop the rest of the data points. That is the code below.

```
def FilteringByYear(df, year_range=(2014, 2024)):
    # Filter the DataFrame for rows within the year range
    filtered_df = df[df['Year'].between(year_range[0], year_range[1])]

    # Get the most recent year for each entity
    most_recent_df = filtered_df.loc[filtered_df.groupby('Entity')['Year'].idxmax()]

    # Reset the index for better readability
    most_recent_df.reset_index(drop=True, inplace=True)

    return most_recent_df
```

We ended up having to scrap this code because when we checked how many total

datapoints we had was less than 1000. The only filtering we kept is where we found the most recent year in a dataset.

```
df2 = spending_life_expectancy[spending_life_expectancy.Year == spending_life_expectancy.Year.max()]
df2  # This sub-data frame shows the data values for the recent-most year.
```

Merging

Merging the data allowed us to perform the other task more effectively rather than performing the same task on multiple sets of data. We decided to merge the data sets for healthcare expenditure and GDP per capita, so that we could calculate what proportion of gross income was spent on healthcare. We then merged this data set with the Gini Coefficient, based on country and year so that it would be possible to determine how the proportion of income spent on healthcare impacts the life expectancy disparity between sexes. After creating the merged data frame, for ease of handling, we changed the data frame to only include the relevant columns: the country, the year, the healthcare expenditure, the GDP per capita, the GINI coefficient, and the life expectancy. Finally, a calculated column was added, dividing the healthcare expenditure by the GDP per capita to determine the proportion of gross income spent on healthcare. The code is below for this process.

```
#Create merged Dataframe to combine healthcare expenditure and GDP per capita with lifeexpectancy for men and women, based on country and year
health_capita = pd.merge(healthcare_expenditure,gdp_per_capita,how='inner', on=['Entity','Year'])
spending_life_expectancy = pd.merge(health_capita,diff_sexes,how='inner', on=['Entity','Year'])
#Keep only the columns need for graphs
spending_life_expectancy = spending_life_expectancy.loc[:,['Entity','Year','Life expectancy - Sex: female - Age: 0 - Variant: estimates','Life expectancy - Sex: male - Age: 0 - Variant: estimates','Current health expenditure per capita, PPP (current international $)']]
#Add calculated column for what proportion of income is spent on healthcare
spending_life_expectancy['healthcare spending/gdp per capita']=spending_life_expectancy['Current health expenditure per capita, PPP (current international $)']/spending_life_expectancy['GDP per capita']
```

Sorting

To perform the graphing we wanted to, we needed to sort the data frames by continent. This way we could do a much broader overview of life expectancy trends for each of the continents. This is done by defining the function sortByContinent, that when called iterates through a list of all

the possible continents, and outputs a dictionary, where the key is the name of the continent, and the value is a data frame with the rows that were labeled as that continent. t.

The code for this function is below.

```
def sortByContinent(df):  
    continents={}  
    for i in df['Continent'].drop_duplicates():  
        continents[i]=df.loc[df['Continent']==i]  
    return continents
```

Data Visualization

We created 5 diverse types of graphs to analyze and demonstrate the relationship between the key economic indicators and male and female life expectancies.

Visualization 1

Our first visualization was a bar graph that showed the top 5 countries with the highest and lowest Gini coefficient values. The Gini coefficient is measured on a scale from 0 to 1 where 0 indicates perfect equality and 1 indicates perfect inequality. Hence, the Gini coefficient helps us understand the level of lifespan inequality based on sex in each country. Therefore, countries with the lowest Gini coefficient values had the most similar life expectancy for men and women. As predicted, these are countries with higher standards of living, where people live way above the poverty line and enjoy a healthy lifestyle. Furthermore, the countries with the highest Gini coefficient values had the highest amount of inequality for life expectancy based on sex. As predicted, these are countries with low standards of living, where people live below the poverty line and do not have the economic resources to live a healthy life.

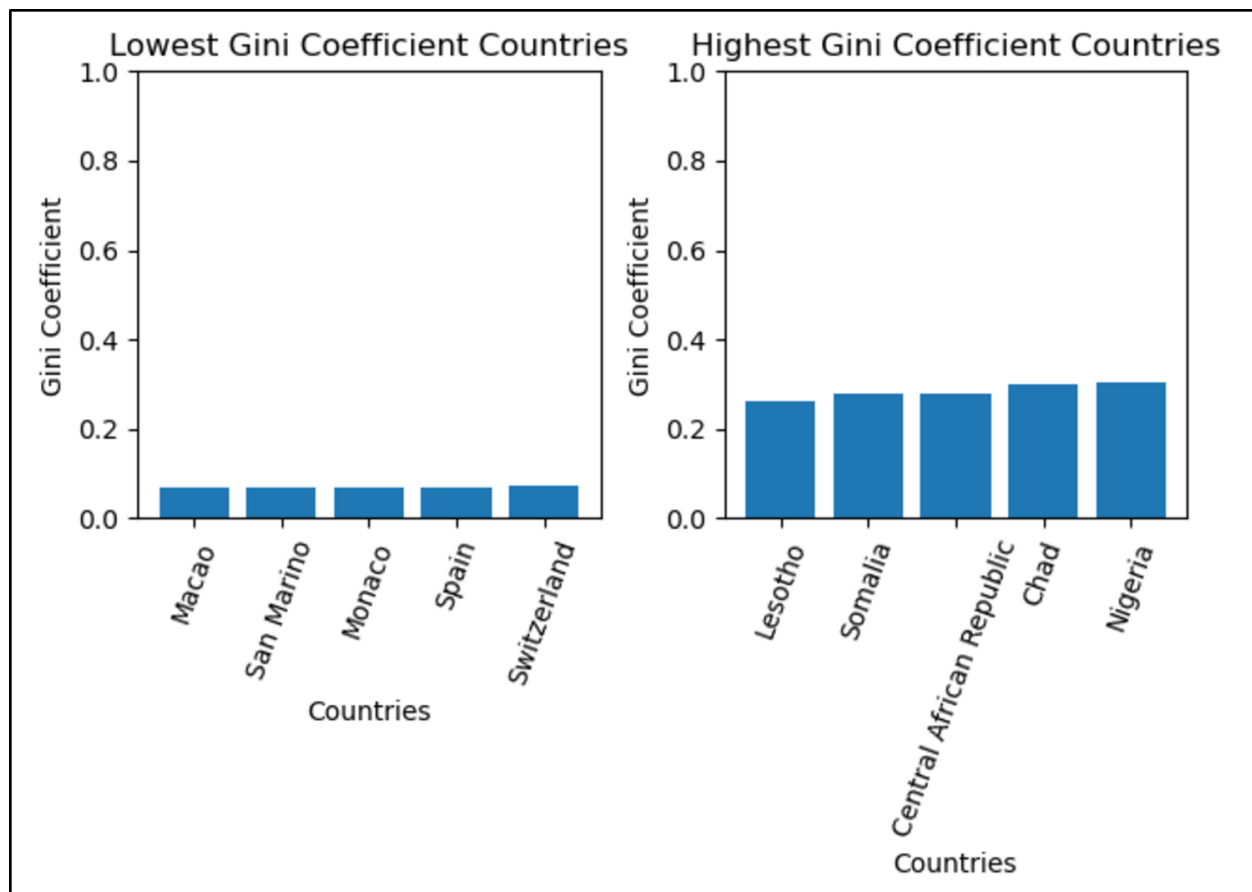
To get this type of visual, I first had to sort the Gini coefficient values in the cleaned dataset in ascending order where the first 5 values were the lowest Gini coefficient values, and the last 5 values were the highest Gini coefficient values. The images below show my code and its corresponding bar graph.

```
fig, ax = plt.subplots(nrows = 1, ncols = 2, sharex = False, sharey = False) # This creates the sub-plots where the first plot will show the top 5 countries with the lowest Gini coefficient and the second plot will show the top 5 countries with the highest Gini coefficient.

ax[0].bar(df1.head().Entity, df1.head()['Gini coefficient of lifespan inequality - Sex: female']) # This creates the first bar graph.
ax[0].set_title("Lowest Gini Coefficient Countries")
ax[0].set_xlabel("Countries")
ax[0].set_ylabel("Gini Coefficient")
ax[0].tick_params(axis = 'x', rotation = 70)
ax[0].set_ylim(0,1) # The y-limit is set from 0 to 1 because the gini coefficient is measured between 0 to 1 where 0 indicates perfect equality and 1 indicates perfect inequality.

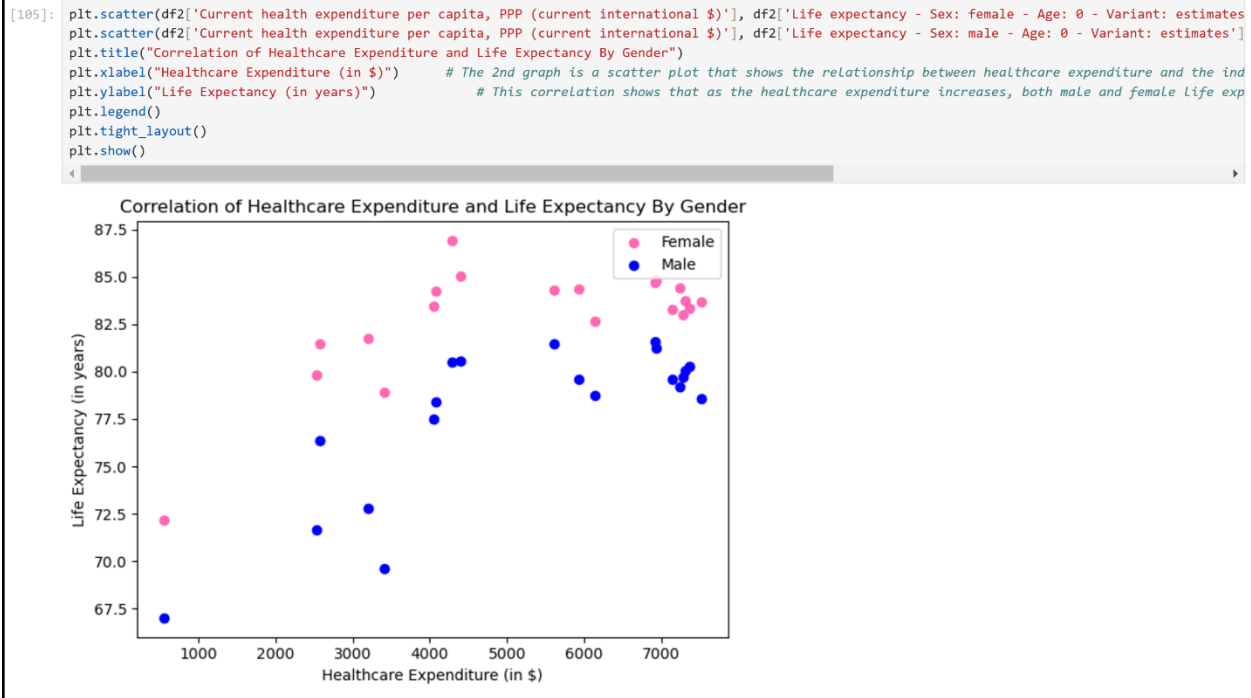
ax[1].bar(df1.tail().Entity, df1.tail()['Gini coefficient of lifespan inequality - Sex: female']) # This creates the second bar graph.
ax[1].set_title("Highest Gini Coefficient Countries")
ax[1].set_xlabel("Countries")
ax[1].set_ylabel("Gini Coefficient")
ax[1].tick_params(axis = 'x', rotation = 70)
ax[1].set_ylim(0,1)

plt.tight_layout()
plt.show()
```



Visualization 2

Our second visualization was a scatter plot that depicted the correlation between healthcare expenditures and male and female life expectancies. To create this type of visual, I first had to filter the cleaned dataset to get a sub-data frame that only consisted of the recent-most year. In our case, the year was 2021. I used this sub-data frame to create the scatter plot. In my graph, the x-axis was the healthcare expenditure measured in dollars, and the y-axis was the life expectancy measured in years. I color-coded the data points such that all the blue points represented male life expectancy, and the pink points represented female life expectancy. (See image below).

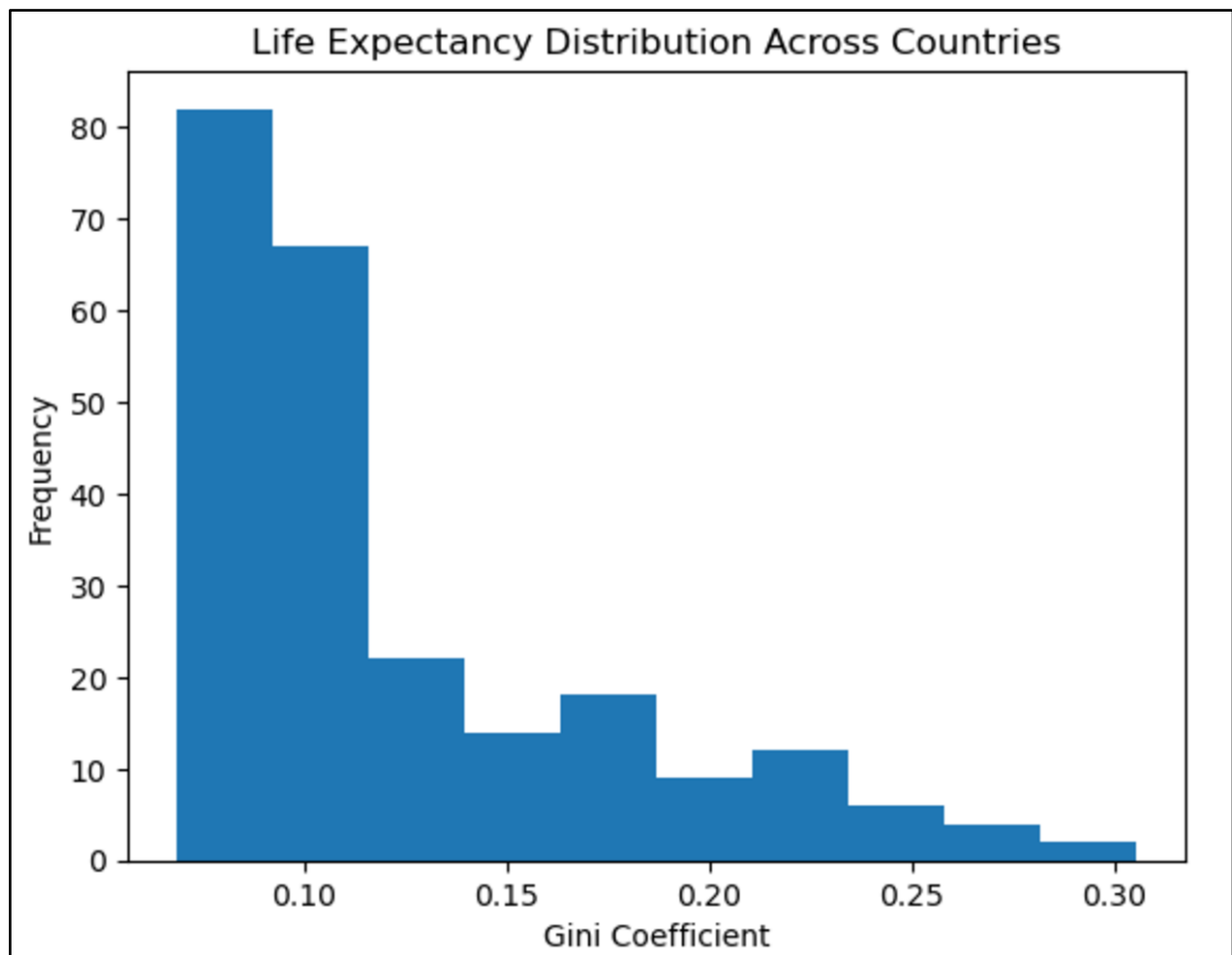


The scatter plot above shows a clear positive correlation between healthcare expenditure and life expectancy. This indicates that as each country spends more on its healthcare system and facilities, the life expectancy of both males and females in that country increases. To be more precise, at any given healthcare expenditure amount, female life expectancy tends to show a sharper increase when compared to male life expectancy. Hence, all the pink points are above the blue points. This can also mean that women tend to prioritize their health more than men and hence, prefer to spend more on healthcare.

Visualization 3

Our third visual was a histogram that showed the inequality of lifespan based on sex distribution across the different countries in our dataset. To get this type of graph, I used all the Gini coefficients from the cleaned dataset. Since I did not filter for a specific year, this histogram also enables us to see the lifespan inequality distribution over the years from 2000 to 2021. In

this histogram, the x-axis represents the distinct categories of Gini coefficients (bins), and the y-axis represents the frequency of each bin. (See image below).

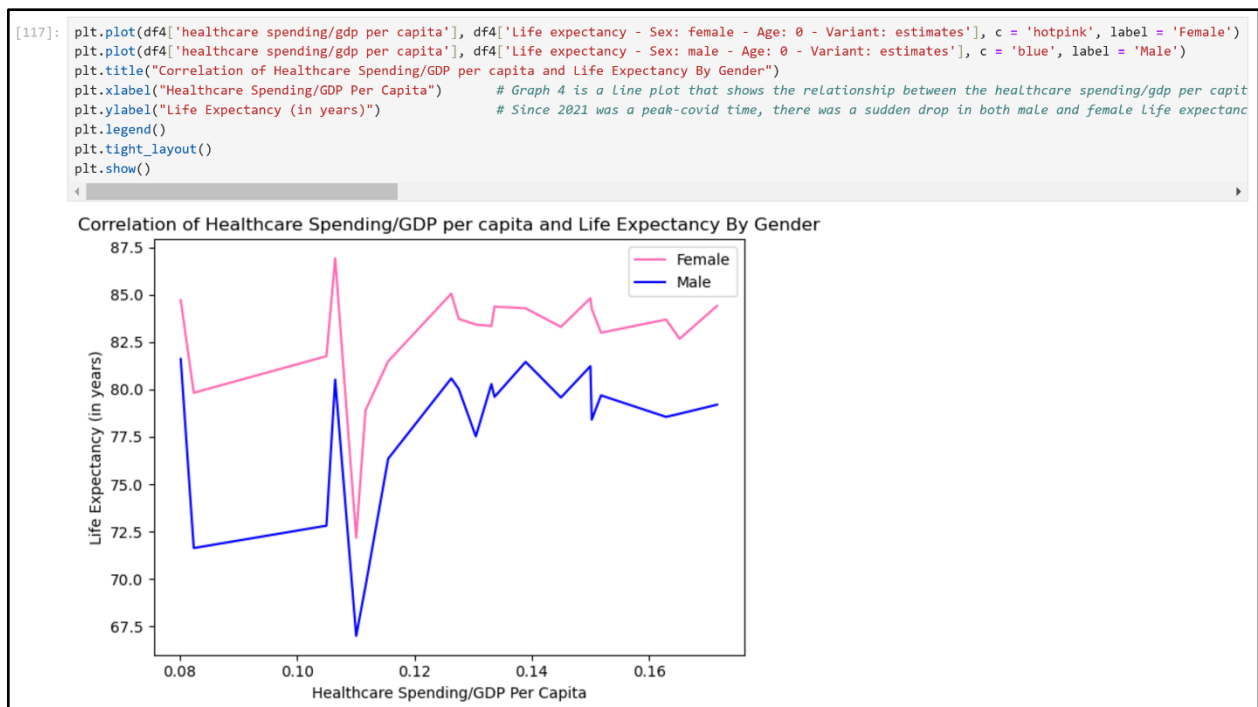


The histogram above shows that most of the countries in our dataset had a Gini coefficient closer to 0.10, indicating that most countries had lower inequality for lifespan.

Visualization 4

Our fourth visualization was a line plot that portrayed the relationship between the healthcare expenditure-GDP per capita ratio and the male and female life expectancy. To get this type of graph, I had to filter the cleaned dataset for the most recent year, 2021. The healthcare expenditure-GDP per capita ratio was created by dividing the healthcare expenditure by the GDP per capita. This ratio indicates how much countries spend on healthcare for income generated per

person in their country. Hence, a ratio number higher than 1 would indicate that the country spends more on healthcare relative to the income generated by each person whereas a ratio number lower than 1 would indicate that the country spends less on healthcare relative to the income generated by each person in that country's economy. (See image below).



The line plot above shows that there is a slight upward trend between the healthcare spending-GDP per capita ratio, starting from when the ratio is at about 0.12, and the male and female life expectancy in years. In this graph, the blue line represents the male life expectancy, and the pink line represents the female life expectancy. There are some exceptions to the trend, but on average the ratio of how much of a country's income is spent on healthcare does not have a significant impact on life expectancy.

Visualization 5

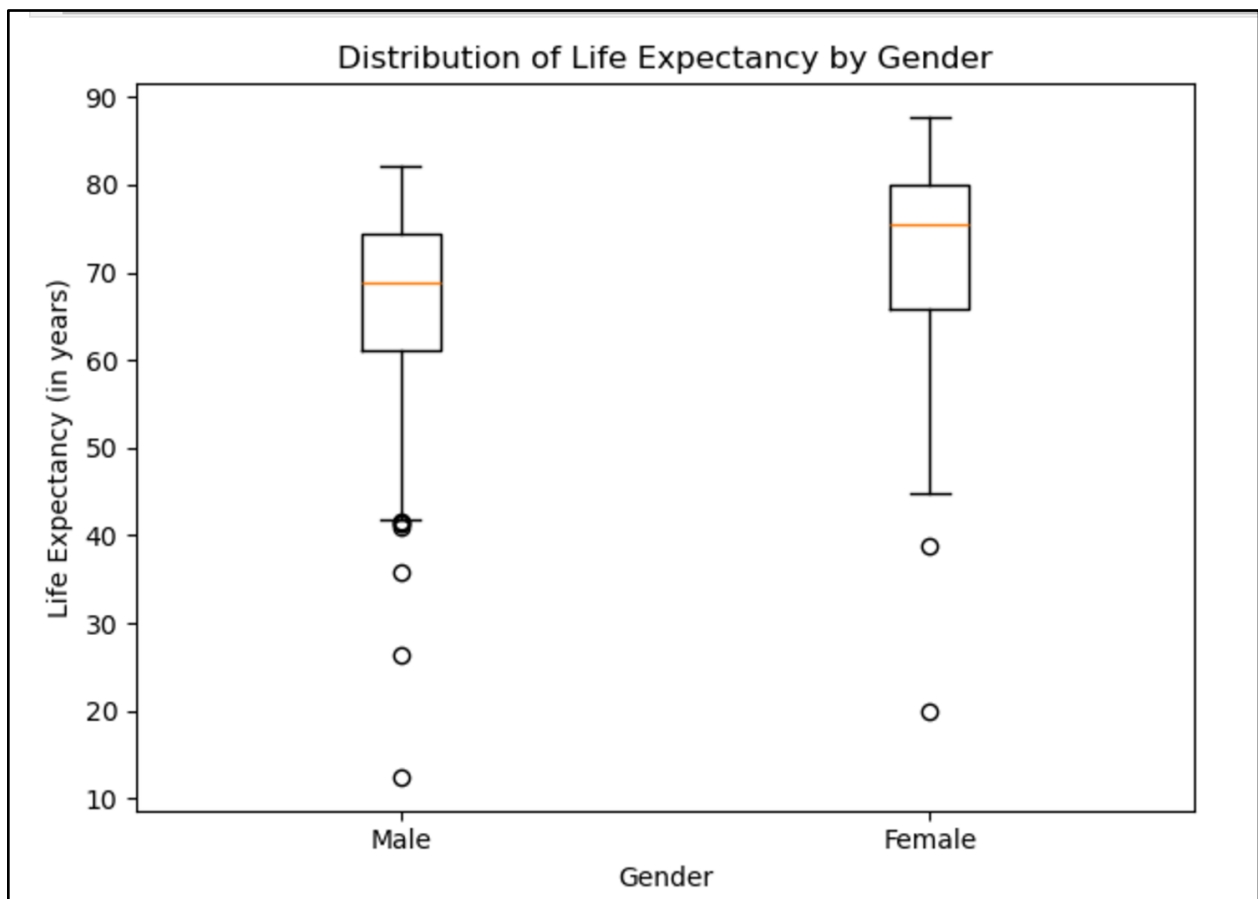
Our fifth and final data visualization was a box plot representing the distribution of the world life expectancy for males and females from 2000 to 2021. To create this type of graph, I

first had to build a separate sub-data frame that had the male and female life expectancy from my cleaned dataset. I then merged these two sub-data frames when inputting them into the box-plot function. (See images below).

```
[121]: male_data = spending_life_expectancy['Life expectancy - Sex: male - Age: 0 - Variant: estimates']
female_data = spending_life_expectancy['Life expectancy - Sex: female - Age: 0 - Variant: estimates']
total_data = [male_data, female_data]

plt.boxplot(x = total_data) # This box plot shows the 5-number summary (min, 25th percentile, median, 75th percentile, max,
plt.title("Distribution of Life Expectancy by Gender") # This box plot shows that women, in general, tend to live longer than men, regardless of nation
plt.xlabel("Gender")
plt.xticks([1,2], ['Male', 'Female'])
plt.ylabel("Life Expectancy (in years)")

plt.tight_layout()
plt.show()
```



The 2 box plots show the 5-number summary (minimum, 25th percentile, median, 75th percentile and maximum values) for the life expectancy of males and females. Although there are outliers for both male and female categories, the females at every level of the box plot tended

to outlive their male counterparts. Hence, females tend to live longer than males around the world, regardless of their nationality.

Data Pivoting

We created 3 different pivot tables that showed our cleaned dataset in a more organized format. Before I created the pivot tables, I had to simplify the names of some columns and transform our datasets into a multi-index. For the first and second pivot tables, I used the same multi-index dataset that was created by merging our 3 initial datasets: healthcare expenditure, GDP per capita and the male and female life expectancy. The 2 columns that I chose as the multi-index for this merged dataset were the year and country. The third pivot table used the 4th dataset that contained the Gini coefficient and the general life expectancy. The 2 columns that I chose as the multi-index for this dataset were the continent and country.

Pivot Table 1

In the first pivot table, I used country as the index, year as the columns and the values were the male and female life expectancy. I also used the 'aggfunc' parameter to get the average of the male and female life expectancy for each year. (See image below).

```
[139]: df1 = spending_life_expectancy.pivot_table(index = 'Country', columns = 'Year', values = ['Male Life Expectancy', 'Female Life Expectancy'], aggfunc = 'mean')
df1 # This pivot table shows the average Life expectancy for males and females from 2000 to 2021 in each of the countries in the dataset.
```

```
[139]:
```

	Year	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	...	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
Country																						
Afghanistan		NaN	NaN	57.442	58.472	59.073	59.588	60.123	60.804	61.428	61.791	...	60.099	60.534	60.495	60.363	60.661	59.388	59.175	59.910	58.455	NaN
Albania		77.926	78.031	78.142	78.329	78.753	79.303	79.943	80.584	81.076	81.282	...	75.660	75.666	75.830	76.281	76.662	76.997	77.448	NaN	NaN	NaN
Algeria		71.944	72.295	72.842	73.027	73.642	73.962	74.283	74.556	74.827	75.074	...	73.493	73.680	73.850	73.992	74.106	74.177	74.255	74.350	71.733	NaN
Angola		49.132	50.047	50.367	52.181	53.074	54.051	54.811	56.041	57.162	58.129	...	56.677	57.404	58.068	58.680	59.237	59.727	60.199	60.609	60.509	NaN
Argentina		77.288	77.642	77.911	77.808	78.177	78.632	78.437	77.925	78.577	78.593	...	72.851	72.963	73.303	73.367	73.044	73.791	73.951	74.163	72.981	NaN
...		...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
Vietnam		77.278	77.673	78.014	78.186	77.807	77.951	78.049	78.152	78.213	78.272	...	68.979	69.042	69.100	69.147	69.153	69.217	69.303	69.414	70.366	NaN
World		68.920	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	69.071	69.323	69.485	69.835	70.055	69.257	NaN
Yemen		64.616	65.098	65.669	66.284	66.731	67.234	67.679	68.121	68.488	68.812	...	65.553	66.028	65.996	63.985	NaN	NaN	NaN	NaN	NaN	NaN
Zambia		47.295	48.105	49.145	50.419	51.097	52.115	53.346	54.470	55.855	56.933	...	56.954	57.623	58.094	58.545	58.919	59.207	59.693	60.349	60.731	NaN
Zimbabwe		NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	54.066	55.397	56.569	57.330	57.940	58.304	58.852	58.830	59.091	NaN

161 rows × 44 columns

Since the above pivot table contained 'NaN' values, I stacked it to remove these values and display the pivot table in a more organized format. (See image below).

[141]:			Female Life Expectancy	Male Life Expectancy
	Country	Year		
	Afghanistan	2002	57.442	54.986
		2003	58.472	55.849
		2004	59.073	56.519
		2005	59.588	56.880
		2006	60.123	56.974
	...	...	...	...
	Zimbabwe	2016	61.401	57.940
		2017	62.016	58.304
		2018	62.741	58.852
		2019	63.049	58.830
		2020	63.747	59.091
	3336 rows × 2 columns			

The above stacked pivot table shows the average male and female life expectancy for each country over the different years. Since the years are in ascending order, we can observe that the average life expectancy for males and females in each country improved as the years passed.

Pivot Table 2

In the second pivot table, I again used country as the index, year as the columns but the values were changed to healthcare expenditure and GDP per capita. Unlike the first pivot table, I

did not create any 'aggfunc' parameter this time because I wanted to show the exact values from the dataset and how these values changed over the past 20 years. (See image below).

```
[143]: df2 = spending_life_expectancy.pivot_table(index = 'Country', columns = 'Year', values = ['Healthcare Expenditure', 'GDP per capita'])
df2 # The 2nd pivot table shows the healthcare expenditure and GDP per capita from the years 2000 to 2021 for each country listed in the dataset.
```

```
[143]:
```

		GDP per capita ...												
	Year	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	...	2012	2013
Country														
Afghanistan		NaN	NaN	796.8166	842.8052	869.03925	964.4081	1057.0966	1259.9967	1319.6074	1557.3207	...	155.38303	178.07903
Albania		4808.4795	5285.8090	5608.9624	6004.6450	6419.87060	6858.4673	7347.3300	7866.1577	8522.1300	8859.4720	...	643.14575	664.90247
Algeria		6834.5537	7218.2456	7814.8384	8590.1780	9182.29400	9969.6230	10385.7790	10974.7420	11475.0020	11907.0210	...	800.91000	790.96990
Angola		2013.6362	2162.4850	2522.6719	2711.9282	3064.29860	3708.7705	4592.3374	5773.5483	6743.7480	7087.6040	...	177.02948	208.89516
Argentina		14368.9420	13651.9810	12094.7720	13088.5650	14183.22600	15344.1570	16490.4730	17891.9430	18520.3050	17328.9770	...	1920.12780	1960.46580
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
Vietnam		2773.1016	2915.6860	3070.6934	3242.2854	3437.66430	3637.6565	3828.9077	4037.2330	4200.2354	4360.0615	...	318.36240	341.31305
World		9903.8600	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN
Yemen		4212.1055	4249.0337	4292.0986	4326.5103	4371.39550	4485.5830	4497.2593	4516.1436	4549.0090	4593.5425	...	169.29358	170.52568
Zambia		1428.5023	1504.4498	1579.0227	1698.5515	1828.30980	1970.0187	2133.5940	2318.6270	2504.7570	2743.3220	...	131.97697	164.48184
Zimbabwe		NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	162.51550	179.32047

161 rows x 44 columns

To remove the 'NaN' values, I stacked the second pivot table as well. (See image below).

[145]:

		GDP per capita	Healthcare Expenditure
Country	Year		
Afghanistan	2002	796.81660	87.465030
	2003	842.80520	86.229300
	2004	869.03925	93.926300
	2005	964.40810	105.835266
	2006	1057.09660	118.180824
...	...	...	...
Zimbabwe	2016	1782.83010	203.482400
	2017	1843.94630	226.177220
	2018	1900.19920	180.091130
	2019	1753.02440	132.068700
	2020	1585.97280	118.127060

3336 rows × 2 columns

From the above stacked pivot table, we can see that the healthcare expenditure and the GDP per capita for each country over the different years mostly increased. However, there is a sudden decrease for both the healthcare expenditure and GDP per capita in 2019 and 2020, as seen in Zimbabwe's case, because of the pandemic, unemployment, and the recession.

Pivot Table 3

For the third and final pivot table, I used the continent as the index, country as the column and the values were the Gini coefficient and the general life expectancy. Similar to the

second pivot table, I did not create any 'aggfunc' parameter because I wanted to show the exact values from the dataset and how these values changed over the past 20 years. (See image below).

[157]: df3 = income_inequality.pivot_table(index = 'Continent', columns = 'Country', values = ['Gini Coefficient', 'General Life Expectancy'])
df3 # This pivot table shows the Gini coefficient and General Life Expectancy for all the countries as columns and continents as the index.

[157]:

	General Life Expectancy ...															
Country	Afghanistan	Albania	Algeria	American Samoa	Andorra	Angola	Anguilla	Antigua and Barbuda	Argentina	Armenia	...	Uruguay	Uzbekistan	Vanuatu	Venezuela	Vietnam
Continent																
Africa	NaN	NaN	75.159	NaN	NaN	61.042	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN
Asia	62.27	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	74.229	...	NaN	0.111182	NaN	NaN	0.108666
Europe	NaN	78.358	NaN	NaN	84.532	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN
North America	NaN	NaN	NaN	NaN	NaN	NaN	77.741	76.892	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN
Oceania	NaN	NaN	NaN	72.654	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN	0.130925	NaN	NaN
South America	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	76.6	NaN	...	0.10353	NaN	NaN	0.104659	NaN

6 rows x 472 columns

Since the above pivot table had a lot of 'Nan' values, when compared to the first and second pivot tables, I stacked the third pivot table as well. (See image below).

[161]:

		General Life Expectancy	Gini Coefficient
Continent	Country		
Africa	Algeria	75.159	0.108450
	Angola	61.042	0.211194
	Benin	59.123	0.237901
	Botswana	63.215	0.178921
	Burkina Faso	58.778	0.230016
...	...	...	...
South America	Paraguay	73.454	0.123706
	Peru	75.732	0.103351
	Suriname	70.567	0.130916
	Uruguay	77.285	0.103530
	Venezuela	73.004	0.104659

236 rows × 2 columns

The stacked pivot table above shows each of the countries sorted by their continents. Furthermore, we can also see that as the Gini coefficient moves closer to 0, the longer the general life expectancy in each country, regardless of the continent that they belong to.

Conclusion

In conclusion, the data visualizations and pivoting effectively demonstrate that all three key economic indicators affect life expectancy.

Visualization 1 helped us identify the top and bottom 5 countries based on the life expectancy disparity based on sex using the Gini coefficient. Visualization 2 showed us the positive correlation between healthcare expenditure and male and female life expectancy.

Visualization 3 showed the level of disparity of life expectancy based on sex across countries.

Visualization 4 showed a slight positive correlation between the healthcare expenditure-GDP per capita ratio and the male and female life expectancy. Visualization 5 helped us understand the nuanced differences between male and female life expectancies for the last 20 years.

Then, pivot table 1 showed the average male and female life expectancy in each country from 2000 to 2021. After stacking this pivot table, we could see that as the years passed, the average male and female life expectancies in each country steadily increased. Pivot table 2 showed the healthcare expenditure and the GDP per capita in each country from 2000 to 2021. After stacking this pivot table, we could observe that both the healthcare expenditure and the GDP per capita in each country increased as the years passed. This makes sense logically, because as the country's income increases, they have more available income to spend on healthcare. Pivot table 3 showed the relationship between the Gini coefficient and the general life expectancy where it was subtly obvious that as the Gini coefficient decreased, the life expectancy increased in each country.

Hence, we can conclude that increased spending on healthcare results in a higher life expectancy for both men and women, as shown by visualization 2. The Gini coefficient is negatively correlated with life expectancy. This means that countries with higher inequality for life expectancy between the sexes, on average tend to have a lower overall life expectancy. Even with these indicators in effect, the female life expectancy tended to be higher than the male life expectancy in each country in the last 20 years.

Works Cited

<https://www.kaggle.com/datasets/mahdiehhajian/life-expectancy-around-the-world?select=life-expectancy-of-women-vs-life-expectancy-of-men.csv>

<https://www.kaggle.com/datasets/mahdiehhajian/life-expectancy-around-the-world?select=life-expectancy-versus-gini-lifespan-inequality.csv>

<https://www.kaggle.com/datasets/mahdiehhajian/life-expectancy-around-the-world?select=life-expectancy-vs-gdp-per-capita.csv>

<https://www.kaggle.com/datasets/mahdiehhajian/life-expectancy-around-the-world?select=life-expectancy-vs-healthcare-expenditure.csv>